

Monthly Intelligence Report

Edition 001 — Alpine Hydropower & Grid Transition

Where the energy transition is outrunning the grid. Inaugural edition · [Month Year] · SSI v4.0.2

CONTENTS

A Executive Summary	E European Context Card
B Cyber & Economic Exposure Monitor	F SSI Enhanced Neural Network
C Data Refresh & Changelog	G Looking Ahead
D Deep-Dive: Alpine Hydropower & Grid Transition	

SECTION A

Executive Summary

SUBSTATIONS SCORED

947
147 HV (≥220 kV) · 776 MV · 24
junction

POWER LINES MAPPED

625
~8,000 km coverage · real OSM
data

MC SIMULATIONS

9.47 M
10,000 per substation

PUBLIC DATA SOURCES

25
95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score. Its engine processes 346,000+ source data points per run, executes 9.47 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and produces full uncertainty quantification across 947 real OSM substations spanning 160 Bezirke, 26 Cantons, and 625 power lines covering approximately 8,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 25 verified public sources — including BFE, ElCom, Swissgrid, Pronovo, BFS, SED, BAFU, Swisstopo, MeteoSwiss, and OpenStreetMap — making the methodology fully auditable and reproducible. Switzerland presents unique grid challenges: approximately 60% of electricity generated from Alpine hydropower, a multi-decade nuclear phase-out under the Energy Strategy 2050, distributed solar integration across diverse topography, and cross-border energy trading at the heart of the European interconnected grid.

Substation in Focus

EKS

Schaffhausen, Schaffhausen · 20 kV

0.388

Medium

0.149

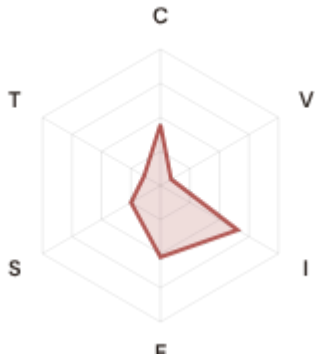
21th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.448 / 0.30 (149%)

I Infrastructure: 0.654 / 0.25 (262%)

S Saturation: 0.250 / 0.20 (125%)

V Voltage: 0.091 / 0.10 (91%)

E Economic: 0.525 / 0.10 (525%)

T Transition: 0.139 / 0.05 (278%)

MODIFIERS

R3 Consequence: 0.886 ↓ dampens

R4 Graph Criticality: 1.042 ↑ amplifies

R6a Restoration: 1.004 → neutral

R6b Seismic: 1.004 → neutral

R7 Digital Readiness: 1.000 → neutral

This month's spotlight — automatically selected for June 2026 — is **EKS** in Schaffhausen, Schaffhausen. With an R_final of 0.388 (Medium band, 21th fleet percentile), its risk profile is dominated by the E (Economic) component at 525% of its weight ceiling, followed by T (Transition) at 278%. Modifiers collectively shift R_base by 6.7%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — STANDING ITEM

Cyber & Economic Exposure Monitor

Why this section exists: Switzerland's grid faces unique vulnerabilities: Alpine terrain isolation limits redundancy, cross-border energy flows create dependency on neighbours, the nuclear phase-out reduces baseload capacity, and a small population means high per-capita economic consequence per substation failure. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross silos — identifying substations where low digital resilience intersects with high economic consequence.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

0

High/Critical + R7 < median

AVG R7 MODIFIER

1.0000

Range [0.99, 1.05]

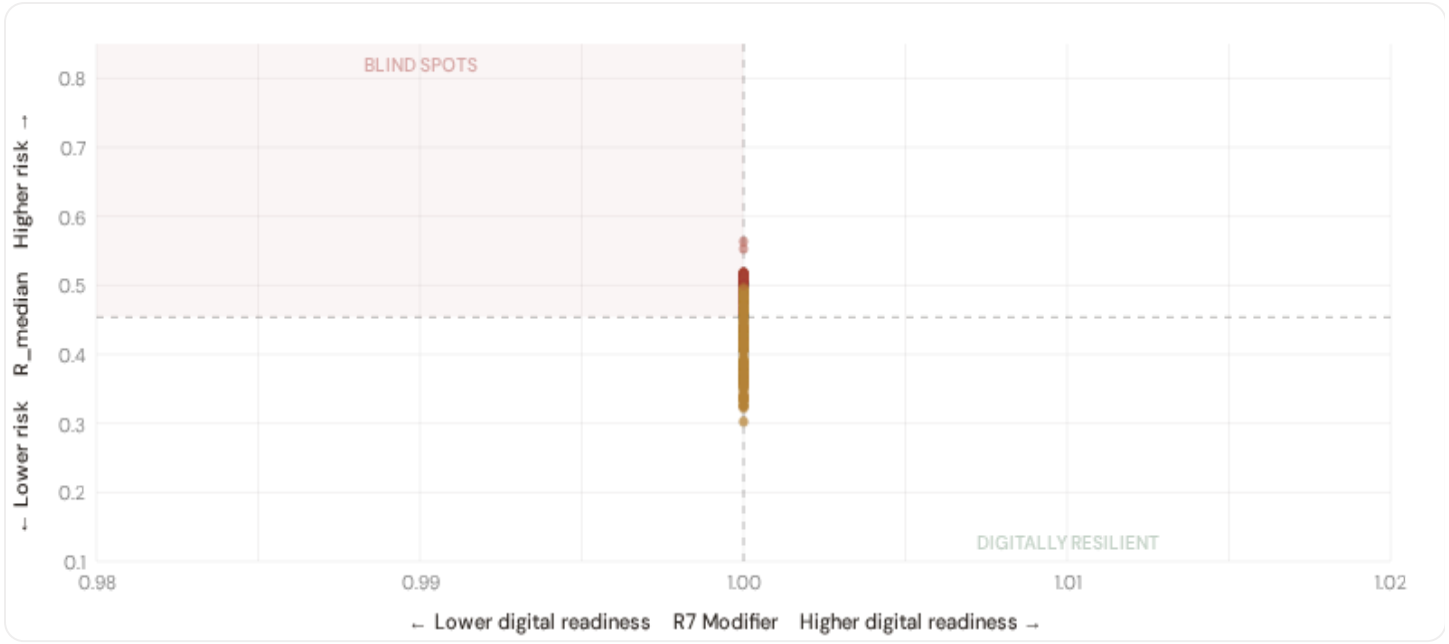
HIGH-CONSEQUENCE SUBS

0

0.0% · R3 ≥ 1.13

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

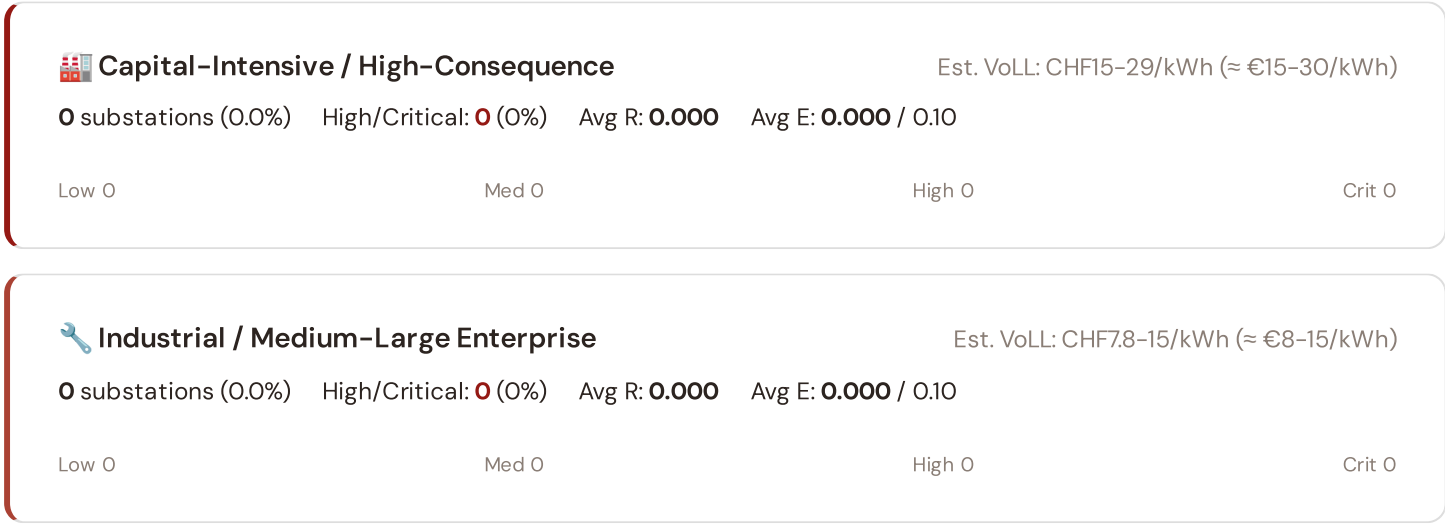


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **0 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0000$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in —. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Commercial / Mixed

Est. VoLL: CHF2.9–7.8/kWh (≈ €3–8/kWh)

0 substations (0.0%) High/Critical: 0 (0%) Avg R: 0.000 Avg E: 0.000 / 0.10

Low 0

Med 0

High 0

Crit 0

Light-Rural / Agricultural

Est. VoLL: CHF1–2.9/kWh (≈ €1–3/kWh)

947 substations (100.0%) High/Critical: 164 (17.3%) Avg R: 0.443 Avg E: 0.508 / 0.10

Low 0

Med 783

High 164

Crit 0

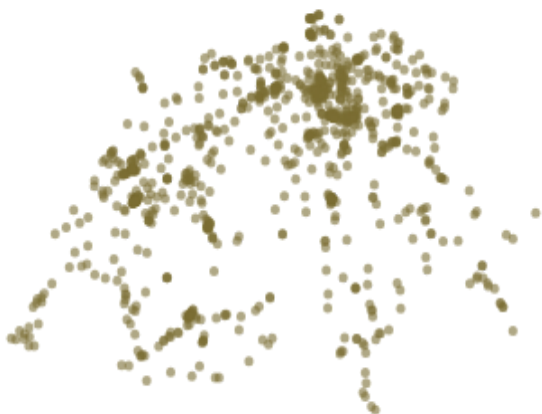
Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **0 substations** combine capital-intensive economic fabric ($R3 \geq 1.14$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in — — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 3.2%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Bezirk Digital Readiness Ranking

Bezirke ranked by average R7 modifier — lower values indicate weaker digital infrastructure. Cross-referenced with average R_median to identify Bezirke combining digital exposure with high grid stress.

R7 Digital Readiness
Low High



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Leventina		1.0000
2	Hochdorf		1.0000
3	Fricktal Δ high R		1.0000
4	Ilanz		1.0000
5	Brugg Δ high R		1.0000
6	Renens Δ high R		1.0000
7	Delemont		1.0000
8	Winterthur Δ high R		1.0000
9	Pinot Δ high R		1.0000
10	Bernina		1.0000
11	Albula		1.0000
12	Surselva		1.0000
13	Sierre		1.0000

14	Conthey		1.0000
15	Martigny		1.0000

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Leventina** has the lowest average R7 (1.0000), while **Franches-Montagnes** leads at 1.0000 — a spread of 0.0000 across the R7 range. **0 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7’s dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0000, median = 1.0000; 0 substations classified HIGH cyber-exposure; 0 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **35 verified public data sources** feeding 95 variables across 6 components and 7 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

35

95 variables

WEEKLY / MONTHLY

4

High-frequency feeds

QUARTERLY / ANNUAL

23

Regular refresh cycle

STATIC / DERIVED

8

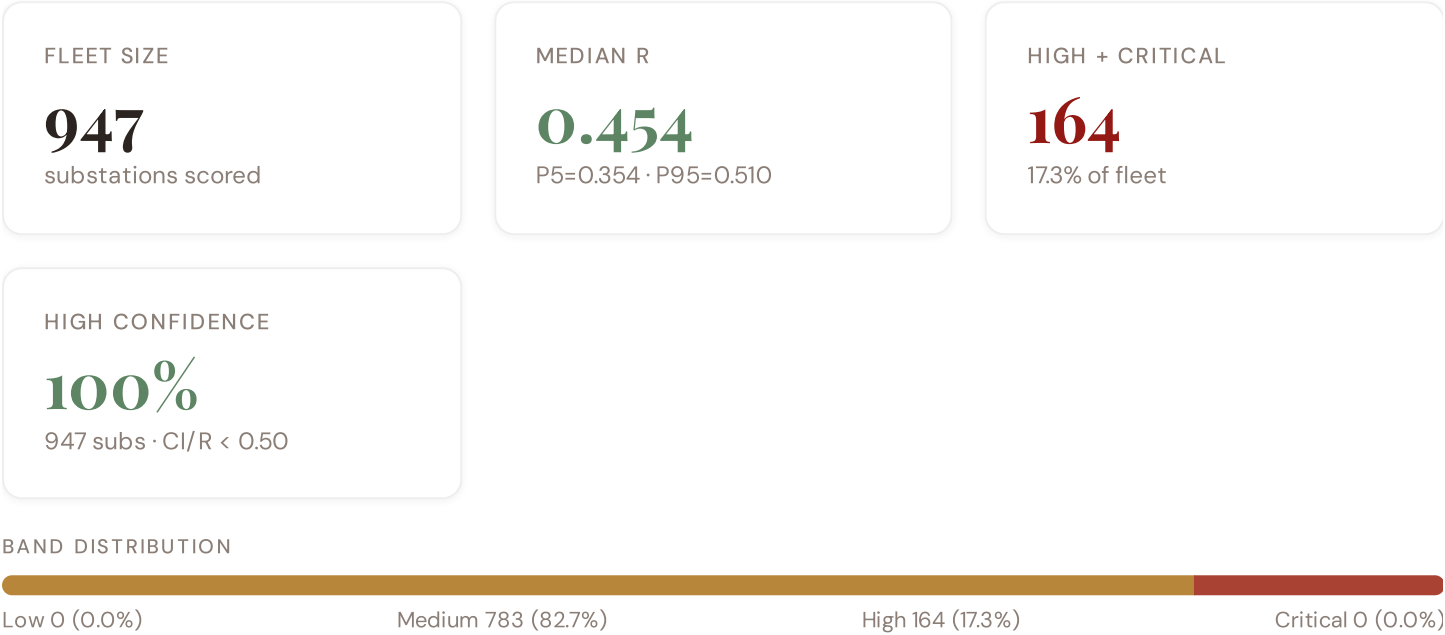
Standards + scenarios

Data Source Registry — June 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
METEO	MeteoSwiss	DAILY	~2 km	3 vars
SGRID	Swissgrid (single TSO)	MONTHLY	Control area	4 vars
PRONOVO	Pronovo (Guarantees of Origin Registry)	MONTHLY	Substation	3 vars
SED	SED Swiss Seismological Service	CONTINUOUS	200+ stations	3 vars
BABS	BABS Civil Protection (Bundesamt)	CONTINUOUS	Kanton	1 var
SECO	SECO State Secretariat for Economic Affairs	QUARTERLY	Kanton	3 vars
SNB	SNB Swiss National Bank	QUARTERLY	Kanton / national	2 vars
ASTRA	ASTRA EV Registry (Bundesamt für Strassen)	QUARTERLY	Kanton	1 var
INNO	Innosuisse — Swiss Innovation Agency	QUARTERLY	Kanton	1 var
KOF	KOF Konjunkturforschungsstelle ETH	QUARTERLY	Regional	2 vars
ELCOM	EICom (Federal Electricity Commission)	ANNUAL	Kanton / DSO	8 vars
BFS	BFS / OFS (Federal Statistical Office)	ANNUAL	Kanton / Gemeinde	8 vars
BAFU	BAFU Federal Office for the Environment	ANNUAL	Kanton / station	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
ICTMIN	ICT-Minimum Standard 2018 (NCSC)	ANNUAL	CH-national	2 vars
NCSC	NCSC National Cyber Security Centre	ANNUAL	CH-national	2 vars
BFE	BFE / OFEN Federal Energy Office	ANNUAL	Kanton	2 vars
ARE	ARE Federal Office for Spatial Development	ANNUAL	Kanton	1 var
EFV	EFV Federal Finance Administration	ANNUAL	Kanton	1 var
VSE	VSE Verband Schweizerischer Elektrizitätsunternehmen	ANNUAL	DSO / Kanton	2 vars

SGRID-N	Swissgrid Network Development Plan	ANNUAL	TSO zone	2 vars
ELCOM-M	EICom Monitoring Report	ANNUAL	DSO-level	2 vars
BFS-S	BFS / OFS Spatial Typology	ANNUAL	Gemeinde	1 var
AVENIR	Avenir Suisse — Policy Research	ANNUAL	Kanton	1 var
AEE	AEE Suisse — Renewable Energy Agency	ANNUAL	Kanton	1 var
COPER	Copernicus CDS / ERA5	STATIC	~25 km	4 vars
SWTOP	swisstopo (Federal Office of Topography)	STATIC	1:25,000	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre (Geosci.)	STATIC	~5 km	2 vars
BFS-GMD	BFS Gemeindeverzeichnis (Municipality Registry)	STATIC	2,148 Gemeinden	1 var
ISO-9223	ISO 9223 Corrosion	DERIVED	Kanton	1 var
ENTSE	ENTSO-E Transparency (CH observer)	HOURLY	CH bidding zone	2 vars
D-0M	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **947 substation scores remain unchanged** this edition. The fleet median R of 0.454 (mean 0.443) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 82.7% in Medium. The first material score changes are expected when the national regulator

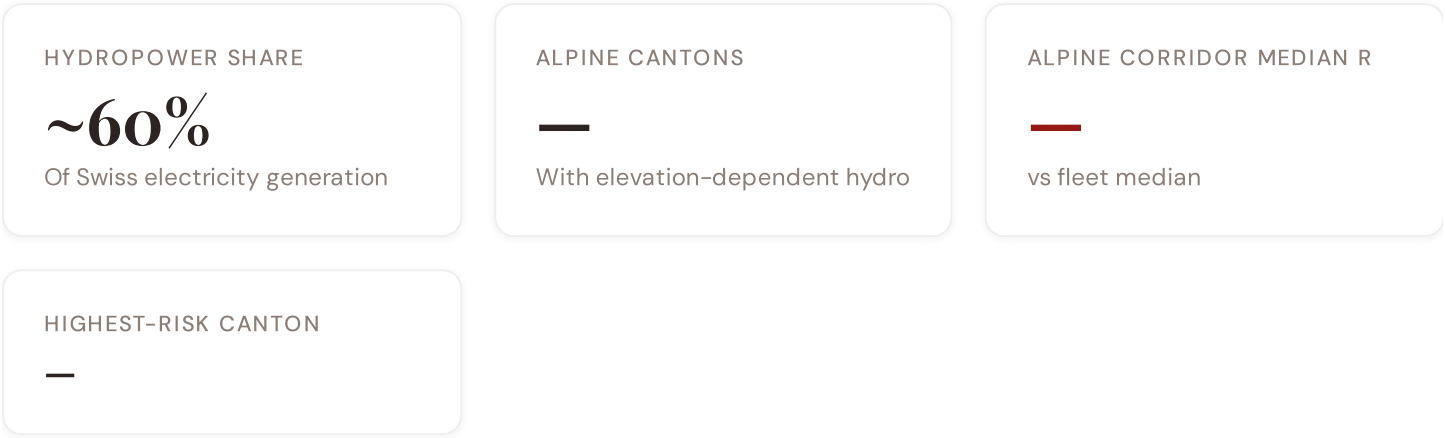
publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

SECTION D — MONTHLY DEEP-DIVE

Alpine Hydropower & Grid Transition: Where Geography Meets the Energy Transition

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Alpine hydropower & grid transition · **002** Nuclear phase-out and baseload replacement · **003** DER saturation in the Swiss Mittelland · **004** Cross-border energy trading and interconnector risk · **005** Voltage quality in Alpine valleys · **006** Infrastructure age vs cantonal investment patterns · **007** Seismic hazard along the Alpine front · **008** T-component forward projections (Energy Strategy 2050+) · **009** Cantonal Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication progress · **012** Year in review.

D.1 Why Alpine Hydropower, Why Now



Loading Alpine corridor data...

D.2 Alpine Corridor Scoring

CANTON	SUBS	MED R	HIGH	C	I	S	T
Loading...							

D.3 Component Analysis — Alpine vs Mittelland

Loading component comparison...

D.4 Implications for the Energy Strategy 2050+

Loading implications...

European Context Card

Switzerland in the European grid: Despite not being an EU member, Switzerland is electrically embedded at the heart of the European interconnected system. Its grid connects to France, Germany, Austria, and Italy through 41 cross-border transmission lines, making it a critical transit corridor for north-south power flows. This integration creates both resilience (import capacity during domestic shortfalls) and vulnerability (exposure to continental frequency events and cascading failures).

CROSS-BORDER LINES

41

To FR, DE, AT, IT

NET IMPORT (TYPICAL)

~5 TWh

Winter dependence on neighbours

SSI COUNTRIES LIVE

4

IT · DE · AT · CH

ENTSO-E ZONE

CH

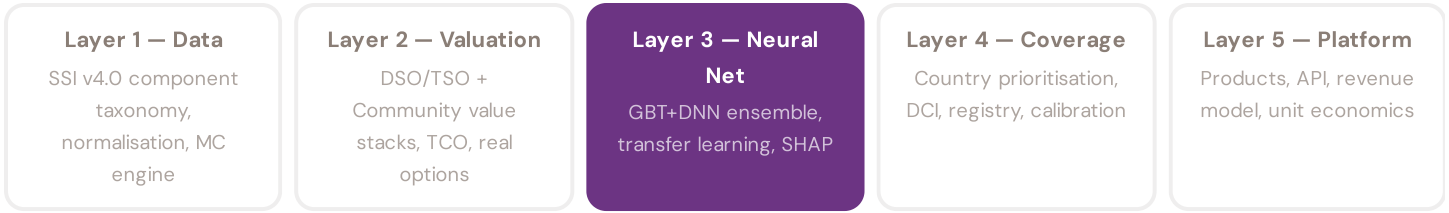
Synchronous Continental Europe

Switzerland's position as a European transit hub means that domestic grid stress cannot be assessed in isolation. The SSI v4.0.2 captures cross-border dependency through several channels: the R3 consequence multiplier reflects the economic importance of interconnection points, the R4 graph criticality modifier identifies substations whose failure would disrupt transit flows, and the E component incorporates import/export dependency into regional economic exposure. With four countries now live in the SSI — Italy, Germany, Austria, and Switzerland — cross-border analysis will become increasingly possible in future editions.

Key European developments relevant to Swiss grid resilience this quarter include: the EU's revised Network Code on Cybersecurity (which Switzerland is expected to adopt bilaterally), ongoing negotiations on a new electricity agreement with the EU, and Swissgrid's participation in the ENTSO-E Winter Outlook process. The absence of a formal EU electricity agreement creates regulatory uncertainty that the SSI's confidence intervals partially capture through wider CI bands in border regions.

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNO and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 947 substations with complete feature vectors. Average CI_width of 0.153 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 783 Medium, 164 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

Edition 002 preview: The next edition will focus on *Nuclear Phase-Out and Baseload Replacement* — analysing how the scheduled decommissioning of Beznau I and II, Gösgen, and Leibstadt affects regional grid stress patterns. The SSI's T-component and I-component will be used to model replacement scenarios and identify substations most exposed to baseload shortfall.

Upcoming data refreshes expected to affect scores include ECom's annual reliability statistics (SAIDI/SAIFI for 2025, expected Q2 2026), Pronovo's updated renewable generation registry, and BFS population/GDP revisions. The SSI pipeline will also incorporate the 2026 SED seismic hazard update, which may shift R6b values for substations along the Alpine front.

Methodological enhancements under development for v4.1 include Bezirk-level OSM topology refinements for improved substation-to-district mapping, live Overpass API integration for infrastructure freshness, and deeper integration of ECom reliability statistics at the cantonal level. The enhanced neural network (ENN) module will expand pattern recognition across the growing multi-country fleet. Switzerland's compact size, high data quality, and transparent regulatory environment make it an ideal validation ground for these enhancements before broader European rollout.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Swiss utility, cantonal energy office, or academic institution and would like to contribute data or methodology feedback, please contact us through the foundation. The SSI's credibility depends on open scrutiny.